User Manual

BLD-305S

(Input voltage 24VDC)

Low-voltage
Brushless DC Motor Driver
with hall sensor

(version: V1.0-0704)

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Read the operating instructions carefully before putting the driver into operation with power

1. Introduction

This series of control drivers uses a closed-loop speed controller and employs high-power MOS electronic component. It utilizes the multiplied frequency signal of brushless DC motor for closed-loop speed control. The control loop has a PID speed regulator, making the system control stable and reliable. The speed control range is 150~20000rpm, and the maximum torque can always be reached at low speed.

2. Features

- PID speed and current dual-loop regulator
- 20kHz PWM frequency
- The overload multiple is greater than 2, and the torque can always reach the maximum at low speed
- Build with over-voltage, under-voltage, over-current, over-temperature, illegal Hall signal and other error alarm functions
- Communication mode adopts standard Modbus protocol. Use two-wire serial link communication based on RS485, transmission mode RTU

3. Specifications

Parameters	BLD-305S
Standard Input Voltage(VDC)	24
Minimum Voltage(VDC)	11.5
Maximum Voltage(VDC)	31
Continuous Output Current(A)	8.0

Note: If using 12VDC power supply, you need to ensure that the voltage is stable. Do not use battery power supply, otherwise the undervoltage protection may be triggered.

4. Terminal Interface

4.1 Power Input

Pin	Pin Name	Description
4	DC+	24VDC input
5	DC-	GND input

4.2 Motor Input

Pin	Pin Name	Description
1	MC	Motor phase C
2	MB	Motor phase B
3	MA	Motor phase A

4.3 Motor Hall input

Pin	Pin Name	Description
1	+5V	Hall signal positive pole
2	HC	Hall signal C phase input
3	HB	Hall signal B phase input
4	HA	Hall signal A phase input
5	GND	Hall signal negative pole

4.4 Signal Control

Pin	Pin Name	Description
1	+5V	+5V Power output port
2	GND	Ground
3	EN	Start / Stop
4	PG	Motor speed pulse output
5	SV	Analogy signal input
6	F/R	Forward / Reverse
7	BK	Brake control
8	ALM	Alarm

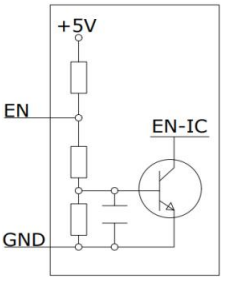
4.5 Control Signal Description:

4.5.1 +5V: Speed control voltage output. You can use a potentiometer to connect SV and GND to continuously regulate the speed of the motor.

4.5.2 GND: Signal ground port.

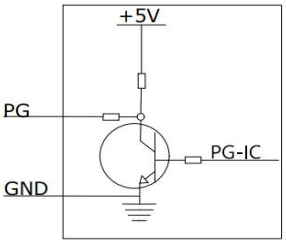
4.5.3 EN: When EN is connected to GND, the motor rotates. When EN is not connected, the motor does not rotate.

(Port diagram)



4.5.4 PG: Motor speed pulse output. The number of pole pairs is P, that is, 3P pulse signals are output per revolution (4 pairs of poles rotate at 1,000 revolutions, and 12,000 pulses are output).

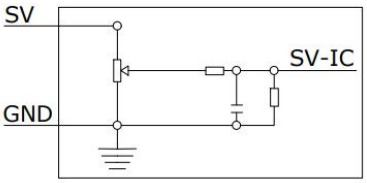
(Port diagram)



4.5.5 SV: Analog voltage input: When connected to an external speed command 0~5V, speed can be adjusted through this port.

The factory-set motor maximum speed supported by the driver is 7000rpm for 2 pairs of poles, 3500RPM for 4 pairs of poles, and 2800RPM for 5 pairs of poles. If you want to change the maximum speed, you need to adjust the position of the jumper cap inside the PCB board.

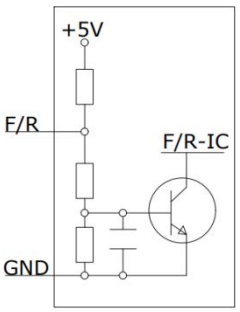
(Port diagram)



4.5.6 F/R: Forward/Reverse control. When it is connected to GND, the motor will rotate reversely, and when disconnected, the motor will rotate forward.

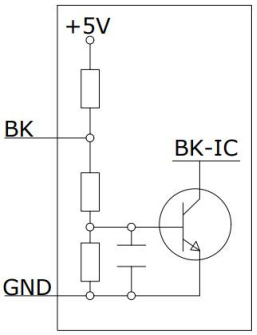
Note: the EN must be turned off first, If you want to switch the forward/reverse direction of the motor .

(Port diagram)



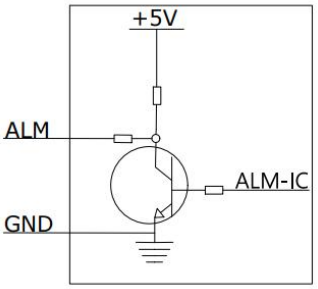
4.5.7 BK: Brake control. When disconnected from GND, the motor operates normally. When connected to GND, the brake operates and the motor stops working. If the load inertia is large, you can adjust the pulse width to get the desired braking effect.

(Port diagram)



4.5.8 ALM: Alarm output. When the driver alarms, this port outputs 0V, and when it does not alarm, it outputs 5V.

(Port diagram)



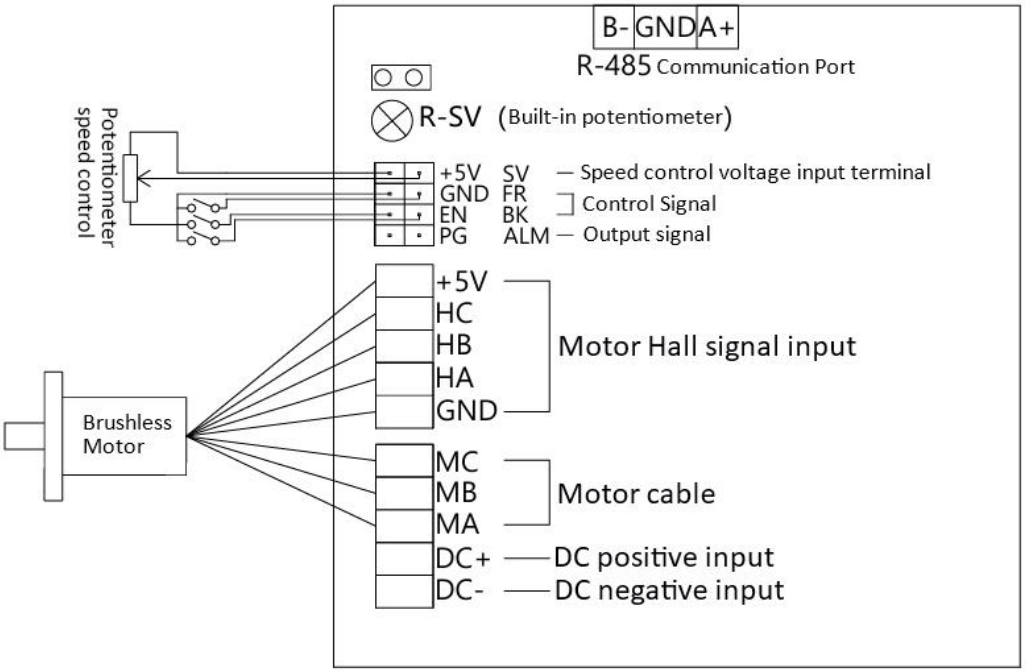
4.5.9 R-SV: Built-in speed limiting potentiometer. It can adjust the motor speed gain (to adjust the speed from 0 to 100%).

4.5.10 PWR/ALM:

Indicator light red: always on after power on, flashing when alarm occurs, different flash times correspond to different alarm states.

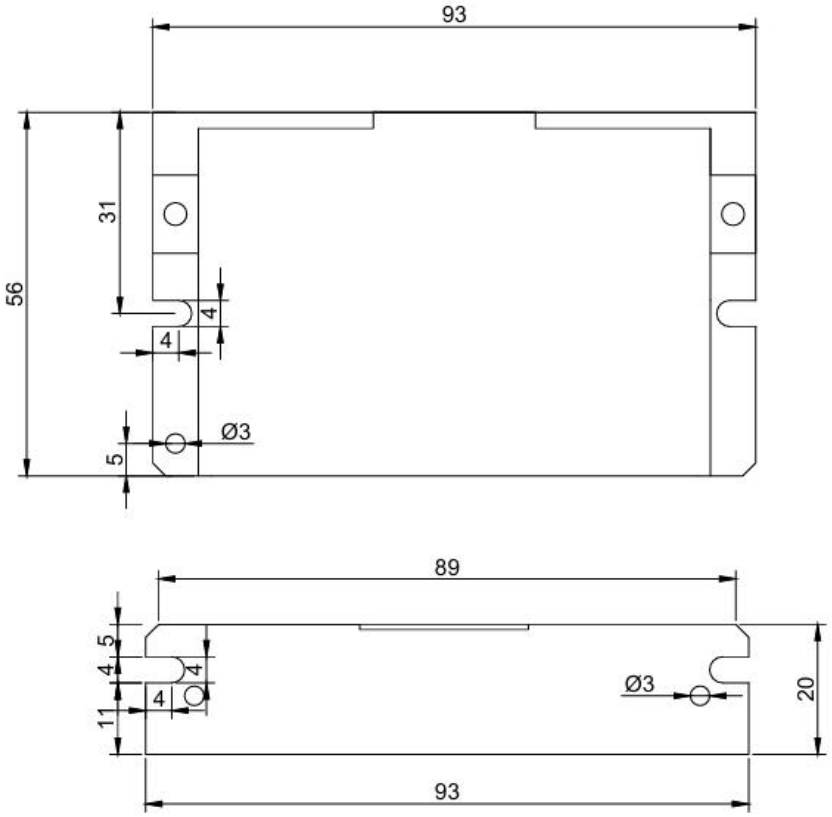
Indicator light green: always on during operation

5. Wiring Diagram of Driver and Motor:



6. Mechanical Specification

(Unit: mm [1inch=25.4mm])



7. Functions and Usage

7.1 Speed Adjustment Method(SV)

External input speed regulation: Connect the two fixed terminals of an external potentiometer to the GND and +5v terminals of the driver respectively, and connect the adjustment terminal to the SV terminal, then you can use the external potentiometer (10K~50K) to adjust the speed.

Analog speed regulation: You can input analog voltage to the SV terminal through other control units (such as PLC, microcontroller, etc.) to achieve speed regulation (relative to GND). The voltage acceptance range of the SV port is DC OV ~ +5V, and the corresponding motor speed is 0 ~ rated speed.

7.2 Motor Operate/Stop Control (EN)

Control the start/stop of the motor by turning on/off the control ports EN and GND. When the port is turned on, the motor runs, otherwise the motor stops. When the EN port is used to control the motor to stop, the motor stops naturally, and its movement pattern is related to the load inertia.

7.3 Motor Rotation Direction Control (F/R)

The running direction of the motor can be controlled by turning on/off the control port F/R and GND. When F/R is disconnected from GND, the motor runs clockwise (facing the motor shaft), otherwise the motor runs counterclockwise.

7.4 Driver Failure (ALM)

When overvoltage or overcurrent occurs inside the driver, the driver will enter a protection state and automatically stop working. The motor will also stop rotating and the blue light on the driver will go out. Only by resetting the enable terminal (i.e. disconnecting EN and GND) or cutting off the power can the driver clear the alarm. If this fault occurs, please check the motor wiring or motor load.

When the driver alarms, the red light of the indicator light will flash. The following is the meaning of the red light flashing frequency:

The red light flashes once time	The motor is stalled
The red light flashes twice times	Over-current alarm
The red light flashes three times	Motor Hall failure
The red light flashes four times	Power supply undervoltage
The red light flashes five times	Power supply overvoltage
The red light flashes six times	Peak current occurs
The red light flashes seven times	The temperature reaches 85℃

8. Communication Method

The communication mode adopts the standard Modbus protocol and uses a two-wire serial link communication based on RS485. The physical interface uses conventional 3-pin 2.54 wiring port (A+, GND, B-) terminals, and serial connection is very convenient. Transmission mode RTU, check mode CRC, CRC start word FFFFH. Data mode: 8-bit asynchronous serial, 1 stop bit, no parity bit, supports communication rate of 4800-256000.

Function parameters support 03H multi-register reading, 06H single register writing.

Site addresses are as follows:

00: broadcast address

1-250: user address

251-255: special address, users can not use.

Register Address	Number of Registers	Register Value Range	Function Code	Function Description	Default Value
0X0056	1	0-4000	0X06	Motor speed setting	5DCH
0X0066	1	0	0X06	Motor stops	00H
0X0066	1	1	0X06	Motor forward rotates	00H
0X0066	1	2	0X06	Motor reverse rotates	00H
0X0066	1	3	0X06	Motor stops braking	00H
0X0076	1	0	0X06	Clear fault code	00H
0X0086	1	1-100	0X06	The overcurrent time setting (S) coefficient is 0.1	1EH
0X00A6	1	1-247	0X06	Drive address	01H
0X00D6	1	1-255	0X06	Starting torque No sense initial speed	18H 04H
0X00E6	1	1-255	0X06	Acceleration time	00H
0X00E6	1	1-255	0X06	Deceleration time	00H
0X00F6	1	00H=4800 01H=9600 02H=14400 03H=19200 04H=38400 05H=56000 06H=57600 07H=115200 08H=128000 09H=230400 10H=256000 If it exceeds 10H, the baud rate will be 9600.	0X06	Baud rate switching	01H
0X0106	1	0-1	0X06	Auto restart function on/off	00 is off 01 is on
0X0116		1-255	0X06	Number of motor pole pairs	05H
0X0126	1	1-32	0X06	Maximum continuous protection current	2DH 8.5A alarm value
0X0136	1	0-1	0X06	Internal and external modes on/off	01 is internal mode 00 is external mode
0X005F	1		0X03	Actual motor speed	
0X0066	1		0X03	Motor running status 0 - Stop 1 - Forward rotation 2 - Reverse rotation 3 - Stop brake	

0X0076	1		0X03	Fault code 1-Overcurrent 2-Over temperature 3-Overpressure 4-Undervoltage 5-Sensor abnormality 6-Overspeed 8-Stalled 9-Peak current	
0X0086	1		0X03	Over-current time protection coefficient(0.1)	
0X0096	1		0X03	Driver internal temperature	
0X00A6	1		0X03	Driver address	
0X00B6	1		0X03	Actual motor current	
0X00D6	1		0X03	Starting torque No sense initial speed	
0X00E6	1		0X03	Actual acceleration and deceleration	
0X00F6	1		0X03	Baud rate switching	
0X0106	1		0X03	Auto restart function on/of	
0X0116	1		0X03	Number of motor pole pairs	
0X0126	1		0X03	Maximum continuous protection current	
0X00C6	1		0X03	Actual motor voltage	

Note: After modifying the parameters, you need to write 55AAH to the register address $80FF so that the system can save the modified parameters. Otherwise, the default parameters will be restored after power failure!

8.1 Modbus example

Internal mode	01	06	01	36	00	01	A9	F8
Start	01	06	00	66	00	01	A8	15
Stop	01	06	00	66	00	00	69	D5
Reverse Rotation	01	06	00	66	00	02	E8	14
Braking	01	06	00	66	00	03	29	D4
Speed Setting 1000RPM	01	06	00	56	03	E8	69	64
Clear Fault	01	06	00	76	00	00	68	10

8.2 Example of How To Read

8.2.1 Speed query (taking 3000RPM as an example)

Send	01	03	00	5F	00	01	B4	18
Return	01	03	02	0B	B8	B8	44	
0BB8H=3000 Calculation formula: 3000*communication setting pole pair number/actual motor pole pair number=actual speed, that is, the actual motor speed is 3000RPM								

8.2.2 Query the internal temperature of the driver (taking 33°C as an example)

Send	01	03	00	96	00	01	64	26
Return	01	03	02	00	21	78	5C	
21H=33℃ (hexadecimal 21 converted to decimal is 33, indicating that the internal temperature of the driver is 33 degrees Celsius)								

8.2.3 Query the actual current (taking 3A as an example)

Send	01	03	00	B6	00	01	65	EC
Return	01	03	02	00	4D	B8	44	
4DH=77 (the hexadecimal 4D converted to decimal is 77. 25 values are 1A, so the actual current is 77/25≈3A)								

8.2.4 Query the actual voltage (taking 23V as an example)

Send	01	03	00	C6	00	01	64	37
Return	01	03	02	00	5D	79	BD	
5DH=93 (hexadecimal 5D converted to decimal is 93, so the actual voltage is 93/4≈23V)								

8.2.5 Query the motor running status (taking forward rotation as an example)

Send	01	03	00	66	00	01	64	15
Return	01	03	02	00	01	79	84	
01H is the code of the motor to rotate forward								

8.2.6 Query drive address

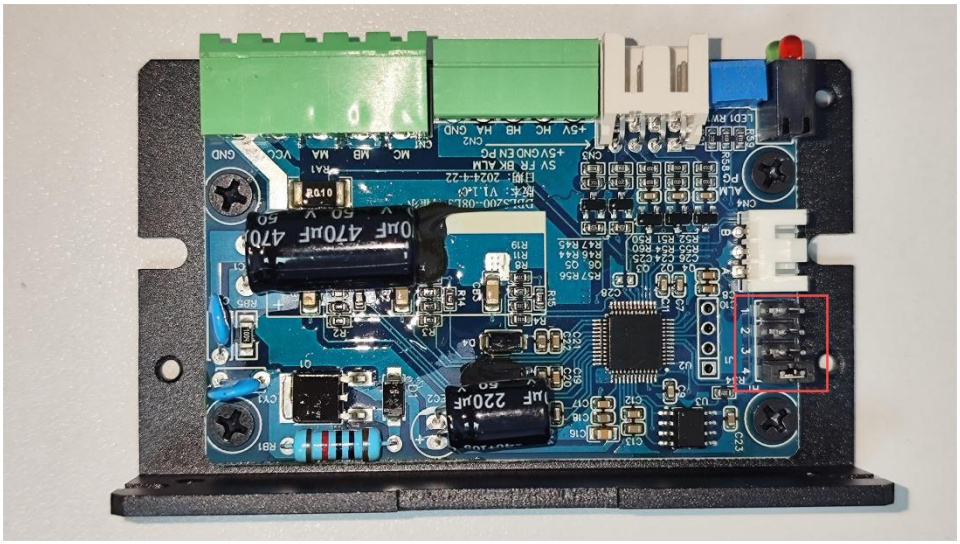
Send	01	03	00	A6	00	01	64	29
Return	01	03	02	00	01	79	84	
01H represents drive address 01								

8.2.7 Query fault

Send	01	03	00	76	00	01	65	D0
Return	01	03	02	00	01	79	84	
01H is the overcurrent alarm code								

9. Brushless DC Motor Maximum Speed Setting

There are jumper caps inside the brushless driver that can be used to set the maximum speed of different motors, as follows: In the picture, if there is a jumper cap at positions 1, 2, 3, and 4, it is marked as 1, and if there is no jumper cap, it is marked as 0. When adjusting the jumper cap, you need to first disassemble the driver shell.



Pin 1	Pin 2	Pin 3	Pin 4	2 pole pair speed(RPM)	4 pole pair speed(RPM)	5 pole pair speed(RPM)
1	0	0	0	1500	750	600
0	1	0	0	1600	1250	1000
1	1	0	0	2500	1080	1350
0	0	1	0	4500	2250	1800
1	0	1	0	5600	2800	2240
0	1	1	0	6250	3120	2500
1	1	1	0	default value 7000	default value 3500	default value 2800
0	0	0	1	8750	4370	3500
1	0	0	1	10000	5000	4000
0	1	0	1	11250	5620	4500
1	1	0	1	12500	62500	5000
0	0	1	1	15000	7500	6000
1	0	1	1	17500	8750	7000
0	1	1	1	20000	10000	8000
1	1	1	1	30000	15000	12000
0	0	0	0	Open Loop	Open Loop	Open Loop